

TRAILGUIDE: AN INFRASTRUCTURE-INDEPENDENT SENSOR-FUSION PEDESTRIAN DEAD RECKONING AND BLE MESH DELAY-TOLERANT FRAMEWORK FOR OFF-GRID SEARCH AND RESCUE

PROJECT REPORT SUBMITTED IN PARTIAL FULFILLMENT OF THE
REQUIREMENTS FOR THE AWARD OF THE
DEGREE OF **BACHELOR OF TECHNOLOGY**

PHASE : I
Nov/Dec 2026

IN INFORMATION TECHNOLOGY
OF THE ANNA UNIVERSITY



PROJECT WORK

Submitted by

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BATCH
2023 – 2027

Under the Guidance of

Dr. D. Saranya, M.E., Ph.D.
SUPERVISOR
Department of Information Technology



Sri Eshwar College of Engineering
(An Autonomous Institution)

Kinathukadavu (Tk), Coimbatore - 641 202, Tamil Nadu Approved
by AICTE, New Delhi and Affiliated to Anna University, Chennai





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BONAFIDE CERTIFICATE

Certified that this Report titled **“TRAILGUIDE: AN INFRASTRUCTURE-INDEPENDENT SENSOR-FUSION PEDESTRIAN DEAD RECKONING AND BLE MESH DELAY-TOLERANT FRAMEWORK FOR OFF-GRID SEARCH AND RESCUE”** is the bonafide work of

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Submitted for the **Autonomous Semester End Project – Phase I Viva-Voce** held on

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INTERNAL EXAMINER

EXTERNAL EXAMINER



Sri Eshwar
College of Engineering
Coimbatore | Tamilnadu
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Affiliated to Anna University, Chennai



QUALITY POLICY

To establish a system of Quality Enhancement, which would on a continuous basis evaluate and enhance the quality of teaching – learning, research and extension activities of the institution, leading to improvements in all processes, enabling the institution to attain excellence.

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To be recognized as a premier institution, grooming students into globally acknowledged engineering professionals.

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- Nurturing research and entrepreneurial culture
- Enabling students to be industry ready and fulfill their career aspirations
- Grooming students through behavioral and leadership training programs
- Making students socially responsible

DEPARTMENT OF INFORMATION TECHNOLOGY

DEPARTMENT VISION

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DEPARTMENT MISSION

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- Empower the faculty to be highly qualified and competent
- Build strong connectivity with various stakeholders to enrich the knowledge
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- Develop and upgrade the facilities for the efficient execution of academic and research activities

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PEO2: Graduates will engage in a post graduate program in the field of Information Technology and Management science leading to academic and research careers

PEO3: Graduates will take up Entrepreneurship as a career.

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PO1: Engineering Knowledge: Apply knowledge of mathematics, natural science, computing, engineering fundamentals and an engineering specialization as specified in WK1 to WK4 respectively to develop to the solution of complex engineering problems.

PO2: Problem Analysis: Identify, formulate, review research literature and analyze complex engineering problems reaching substantiated conclusions with consideration for sustainable development. (WK1 to WK4)

PO3: Design/Development of Solutions: Design creative solutions for complex engineering problems and design/develop systems/components/processes to meet identified needs with consideration for the public health and safety, whole-life cost, net zero carbon, culture, society and environment as required. (WK5) **PO4: Conduct Investigations of Complex Problems:** Conduct investigations of complex engineering problems using research-based knowledge including design of experiments, modelling, analysis & interpretation of data to provide valid conclusions. (WK8).

PO5: Engineering Tool Usage: Create, select and apply appropriate techniques, resources and modern engineering & IT tools, including prediction and modelling recognizing their limitations to solve complex engineering problems. (WK2 and WK6)

PO6: The Engineer and The World: Analyze and evaluate societal and environmental aspects while solving complex engineering problems for its impact on sustainability with reference to economy, health, safety, legal framework, culture and environment. (WK1, WK5, and WK7).

PO7: Ethics: Apply ethical principles and commit to professional ethics, human values, diversity and inclusion; adhere to national & international laws. (WK9)

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PO9: Communication: Communicate effectively and inclusively within the engineering community and society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations considering cultural, language, and learning differences

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PO11: Life-Long Learning: Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change. (WK8)

PROGRAM SPECIFIC OUTCOME

PSO1: Demonstrate the ability to apply knowledge and skills in the development and deployment of software projects.

PSO2: Create innovative solutions by leveraging emerging technologies to effectively manage IT infrastructure.

DECLARATION

We,	
AKASH DHANKAR	722823205003
LINGESH V	722823205027
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Declare that the project entitled “**TRAILGUIDE: AN INFRASTRUCTURE-INDEPENDENT SENSOR-FUSION PEDESTRIAN DEAD RECKONING AND BLE MESH DELAY-TOLERANT FRAMEWORK FOR OFF-GRID SEARCH AND RESCUE**”, submitted in partial fulfilment to Anna University as the project work of Bachelor of Technology (Information Technology) Degree, is a record of original work done by us under the supervision and guidance of Dr. D. Saranya, M.E., Ph.D., Assistant Professor, Department of Information Technology, Sri Eshwar College of Engineering, Coimbatore.

Place: Coimbatore

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ACKNOWLEDGEMENT

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ABSTRACT

Global Navigation Satellite System (GNSS) signal degradation under dense forest canopies presents a critical life-safety hazard to wilderness travelers and Search and Rescue (SAR) teams. Under thick multi-layered foliage, consumer smartphone satellite receivers experience severe Position Dilution of Precision ($\text{PDOP} > 6.5$) or complete signal blackouts. Simultaneously, unassisted Inertial Navigation System (INS) dead-reckoning methods accumulate unbounded quadratic integration drift exceeding 12.4 meters within minutes. Traditional commercial navigation platforms exacerbate these limitations by requiring massive, multi-gigabyte raster map downloads and constant cellular connectivity.

This project presents TrailGuide, an infrastructure-independent wilderness travel and emergency coordination framework engineered for commodity smartphones. TrailGuide introduces a battery-efficient Sensor-Fusion Pedestrian Dead Reckoning (SF-PDR) engine combining accelerometer step kinematics, dynamic Weinberg stride estimation, and a tilt-compensated complementary heading filter fusing gyroscopic angular velocities with magnetometer field vectors. When canopy occlusion causes satellite accuracy to drop below 6.5 meters, the system automatically decouples GNSS updates and propagates spatial coordinates through passive inertial dead-reckoning.

For off-grid emergency signaling, TrailGuide integrates an infrastructure-less background Bluetooth Low Energy (BLE) Delay-Tolerant Network (DTN). Using a controlled Spray-and-Wait protocol with local Hive database deduplication, 24-byte distress beacons are relayed peer-to-peer across mobile human nodes until reaching an internet-connected gateway. Relayed packets are ingested by a cloud-based Next.js and MongoDB Atlas SAR command center featuring 2DSphere spatial query perimeters, live incident status management, and Open Source Routing Machine (OSRM) dynamic road routing.

Field benchmarking across a 2.5 km dense forest trail demonstrates that TrailGuide achieves a mean horizontal localization error of 0.68 meters (94.6% sub-meter accuracy), outperforming raw smartphone GNSS (12.42 m drift) by 94.5% and standard GNSS+INS baselines (4.85 m drift) by 86.0%. The BLE mesh DTN delivers 100% of SOS alerts within 4.2 seconds across 5 hops, while consuming only 11.4% CPU capacity and 13.8% battery per hour during continuous operation.

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LIST OF ABBREVIATIONS

ABBREVIATION	EXPANSION & DESCRIPTION
BLE	Bluetooth Low Energy (2.4 GHz ISM wireless radio standard)
GNSS	Global Navigation Satellite System (GPS, GLONASS, Galileo, BeiDou)
GPS	Global Positioning System (US Satellite Navigation Constellation)
INS	Inertial Navigation System (Strapdown accelerometer/gyroscope system)
IMU	Inertial Measurement Unit (Tri-axial multi-sensor hardware unit)
PDR	Pedestrian Dead Reckoning (Step-stride kinematic positioning method)
SF-PDR	Sensor-Fusion Pedestrian Dead Reckoning (Proposed tracking framework)
DTN	Delay-Tolerant Network (Store-and-forward opportunistic network architecture)
SAR	Search and Rescue (Wilderness emergency operations and extraction teams)
MEMS	Micro-ElectroMechanical Systems (Miniaturized silicon motion sensors)
PDOP	Position Dilution of Precision (Satellite geometric configuration quality factor)
OSM	OpenStreetMap (Open-access collaborative worldwide geographic dataset)
MVT	Mapbox Vector Tiles (Standard protocol buffer encoded spatial vector format)
PBF	Protocol Buffer Format (Binary serialized geographic data encoding)
OSRM	Open Source Routing Machine (High-performance routing and navigation engine)
GATT	Generic Attribute Profile (Bluetooth Low Energy communication layer)
FIFO	First-In, First-Out (Circular sample buffer architecture)
REST	Representational State Transfer (Stateless HTTP web API architectural style)
API	Application Programming Interface (Inter-software communication interface)
UUID	Universally Unique Identifier (128-bit unique packet and beacon identifier)
HUD	Heads-Up Display (Tactical operational high-contrast mobile user interface)

RSSI	Received Signal Strength Indicator (Radio frequency reception power metric)
JSON	JavaScript Object Notation (Lightweight data interchange format)
LoRa	Long Range Low-Power Wide-Area Network Modulation (868/915 MHz chirped spread spectrum)
EKF	Extended Kalman Filter (Non-linear state estimation algorithm)

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Wilderness exploration, eco-tourism, high-altitude mountaineering, backcountry trekking, and Search and Rescue (SAR) missions are conducted in challenging geographic environments that are fundamentally disconnected from conventional communication and computing infrastructure. In remote mountain passes, deep river gorges, densely canopied tropical and temperate forests, and national wildlife reserves, there is a total absence of cellular base transceiver stations (BTS), terrestrial broadband, or Wi-Fi access points. Travelers, forestry field researchers, and rescue personnel who operate in these regions must rely exclusively on mobile consumer equipment for navigation, situational awareness, and emergency signaling.

Modern consumer smartphones are equipped with high-sensitivity multi-constellation satellite receivers capable of processing microwave signals across multiple Global Navigation Satellite System (GNSS) arrays, including the United States Global Positioning System (GPS), Russian GLONASS, European Galileo, and Chinese BeiDou. Under ideal open-sky conditions, these satellite positioning engines achieve horizontal accuracy within 3 to 5 meters. However, when a traveler enters a dense forest canopy, the physical conditions required for satellite localization break down entirely.

Forest vegetation—specifically dense leaf canopies, foliage moisture layers, woody tree trunks, and high-density timber stands—acts as a severe dielectric attenuator of L-band radio frequencies (1.2 GHz to 1.6 GHz). Satellite microwave rays are absorbed, refracted, and scattered across multiple physical obstacles before reaching the smartphone's internal microstrip antenna. This multipath scattering degrades signal timing, introducing pseudorange errors and triggering extreme Position Dilution of Precision ($PDOP > 6.5$). Under multi-layered hardwood and pine canopies, consumer satellite receivers frequently experience complete signal blackout, causing the operating system's location service to freeze the position marker or drift erratically across several hundreds of meters.

Simultaneously, traditional Inertial Navigation Systems (INS) built into smartphones rely on internal Micro-ElectroMechanical Systems (MEMS) accelerometers and gyroscopes. Although inertial dead-reckoning operates independently of external radio signals, unassisted smartphone INS dead-reckoning degrades rapidly over time. Double-integrating raw accelerometer readings to calculate velocity and displacement results in quadratic and cubic error growth driven by sensor bias instability, temperature drift, and thermal noise. In practice, unassisted smartphone dead-reckoning accumulates over 12.4 meters of horizontal drift within minutes of operation.

Furthermore, conventional commercial mobile mapping applications (such as Google Maps, Apple

Maps, AllTrails, and OsmAnd) are poorly suited for off-grid survival. First, their client-server architecture assumes continuous cellular connectivity to stream raster image tiles. When connectivity ceases, un-cached raster viewports go blank. Even when offline caching is manually configured, downloading multi-gigabyte raster image maps exhausts internal device storage, overburdens memory bandwidth, and accelerates battery depletion due to intensive GPU rasterization. Most critically, these commercial applications provide no mechanism to transmit emergency distress beacons without cellular network coverage or expensive, proprietary satellite hardware peripherals such as Garmin inReach, ACR Bivy Stick, or SPOT transceivers.

1.2 PROBLEM STATEMENT

Wilderness travelers, outdoor researchers, and Search and Rescue teams face a critical convergence of technological failures when operating beneath dense forest canopies and remote backcountries:

- **Catastrophic GNSS Attenuation:** Overhead multi-layered foliage introduces severe multipath scattering, signal refraction, and total satellite blackout, causing consumer smartphone positioning errors to exceed 12 to 18 meters.
- **Unbounded Inertial Drift:** Unassisted smartphone INS implementations suffer from rapid quadratic double-integration drift, rendering raw accelerometer tracking unusable without continuous external corrections.
- **Communication Infrastructure Deprivation:** Total absence of cellular networks precludes standard SMS or IP-based emergency SOS transmissions, leaving distressed individuals isolated unless equipped with expensive satellite hardware.
- **Storage and Battery Inefficiency:** Traditional offline mapping platforms rely on monolithic raster tiles consuming hundreds of megabytes per sector and exhausting mobile battery reserves through continuous GPU rendering within 4 to 6 hours.
- **Fragmented Emergency Coordination:** Existing wilderness tools lack seamless integration with centralized SAR dispatch consoles, hindering rapid extraction planning and dynamic overland route navigation.

1.3 OBJECTIVES

The primary objective of this project is to architect, develop, and empirically validate TrailGuide—an infrastructure-independent wilderness travel and emergency coordination platform designed for commodity smartphones. Specific engineering objectives include:

- To engineer a robust Sensor-Fusion Pedestrian Dead Reckoning (SF-PDR) engine combining 100 Hz tri-axial accelerometer kinematics, Weinberg dynamic stride length estimation, and a tilt-compensated complementary heading filter fusing gyroscopic angular velocities with magnetometer magnetic vectors.
- To establish an autonomous dual-state gating mechanism that decouples positioning from degraded satellite feeds when horizontal accuracy drops below 6.5 meters, maintaining sub-meter

dead-reckoning accuracy (mean error < 0.70 m) under continuous canopy blackout.

- To develop an infrastructure-less background Bluetooth Low Energy (BLE) Delay-Tolerant Network (DTN) utilizing a Controlled Spray-and-Wait protocol with local Hive database deduplication to relay 24-byte distress beacons multi-hop across mobile peer nodes.
- To build a lightweight, micro-sharded vector mapping subsystem delivering complete sector trail maps in compact protocol-buffer packages under 2 MB, optimized for low GPU memory overhead and extended battery endurance (>8 hours).
- To create an integrated cloud Search and Rescue (SAR) command center leveraging Next.js, MongoDB Atlas 2DSphere spatial geospatial queries, and Open Source Routing Machine (OSRM) dynamic road routing for real-time extraction coordination.

1.4 SCOPE OF THE PROJECT

The scope of Phase-I encompasses the mathematical modeling, firmware-level sensor acquisition, cross-platform mobile application development, decentralized mesh networking protocols, and full-stack web command room implementation:

- Target Hardware: Commodity iOS and Android smartphones equipped with standard tri-axial MEMS accelerometers, rate gyroscopes, magnetometers, and BLE 4.2/5.0 radios, requiring zero external hardware dongles or proprietary sensors.
- Geographic Environments: Dense forest trails, mountain nature reserves, and university campus woodland paths characterized by complete satellite occlusion and cellular dead zones.
- Communication Protocols: Peer-to-peer opportunistic BLE advertising and scanning adhering to non-connectable undirected advertising standards with binary payload compression.
- Web Command Infrastructure: Modern web technologies including Next.js 14, React Leaflet, MongoDB Atlas geospatial indexing, and containerized OSRM routing instances.

CHAPTER 2

LITERATURE SURVEY

2.1 TRADITIONAL INERTIAL & SATELLITE POSITIONING IN CANOPIES

Keefe et al. (2019) conducted comprehensive empirical evaluations of GNSS performance in commercial timberland forests [7]. Their investigation revealed that canopy closure, basal area density, and tree species composition significantly degrade satellite positioning accuracy. In mature pine and mixed hardwood canopies, positional errors frequently exceeded 10 to 15 meters, with signal acquisition times increasing by an order of magnitude. The authors emphasized the imperative need for complementary sensor integration to support precision forestry and wilderness safety operations.

2.2 GNSS ATTENUATION & MULTIPATH CHALLENGES IN FORESTRY

Bär et al. (2019) investigated multi-sensor fusion architectures for pedestrian micro-navigation across challenging urban canyons and dense temperate forests [18]. Their findings demonstrated that multipath reflections from wet foliage and dense wood trunks introduce non-Gaussian pseudorange measurement errors that severely corrupt standard Extended Kalman Filter (EKF) state estimation. The authors concluded that unassisted satellite positioning is fundamentally unreliable for safety-critical pedestrian tracking under continuous canopy coverage.

2.3 CONVENTIONAL INS & ACCELERATION DOUBLE-INTEGRATION DRIFT

Kumar and Reddy (2020) published an exhaustive review of smartphone-based pedestrian dead reckoning methodologies [15]. Their mathematical analysis demonstrated that double-integrating raw acceleration measurements without kinematic domain constraints leads to quadratic position errors growing as $O(t^2)$. Micro-machined silicon accelerometers exhibit non-zero bias instabilities and thermal drift that accumulate rapidly, producing horizontal positioning errors exceeding 15 meters within 60 seconds of unconstrained integration.

2.4 DELAY-TOLERANT NETWORKS (DTN) IN EMERGENCY COMMUNICATIONS

Gelenbe et al. (2021) analyzed the deployment of Delay-Tolerant Networks (DTNs) in disaster scenarios characterized by total infrastructure collapse [11]. The authors evaluated opportunistic store-carry-and-forward communication paradigms, highlighting that epidemic flooding protocols guarantee high delivery rates but induce severe channel contention and buffer exhaustion. Conversely, quota-based protocols significantly reduce message overhead while maintaining deterministic delivery bounds in sparse human mobility networks.

2.5 OPPORTUNISTIC BLUETOOTH LOW ENERGY (BLE) MESH NETWORKING

Álvarez et al. (2019) designed Bluemergency, an opportunistic communication system leveraging

Bluetooth Mesh and IoT nodes for post-disaster survivor coordination [1]. Their work demonstrated that commodity smartphone BLE radios can establish spontaneous multi-hop mesh topographies without centralized coordinators. Rondón et al. (2019) and Villa et al. (2022) conducted in-depth reliability and latency benchmarks on BLE mesh protocols, concluding that controlled packet flooding over non-connected advertising channels provides resilient emergency beaconing with sub-5-second multi-hop latency [2], [4].

2.6 SENSOR-FUSION & PEDESTRIAN DEAD RECKONING (PDR) TECHNIQUES

Weinberg (2002) introduced a seminal empirical biomechanical stride estimation formulation based on vertical acceleration variation during walking gait cycles [14]. Gallagher et al. (2012) comparatively benchmarked multiple PDR algorithms, establishing that step-and-heading systems constrained by dynamic stride modeling eliminate quadratic acceleration integration drift, reducing error growth to a linear function of step count [19].

2.7 TILT-COMPENSATED ORIENTATION & COMPLEMENTARY FILTERING

Smartphone magnetometers are susceptible to high-frequency electromagnetic anomalies, while gyroscopes suffer from slow low-frequency zero-rate drift. Traditional orientation estimators employ high-order Madgwick or Extended Kalman Filters. However, complementary frequency-domain filtering delivers equivalent heading accuracy with drastically reduced floating-point arithmetic complexity, making it uniquely suited for continuous background execution on low-power mobile processors.

2.8 MACHINE LEARNING & INERTIAL ODOMETRY IN CANOPY ENVIRONMENTS

Recent advances in neural inertial navigation have explored deep learning models—such as RoNIN (Robust Neural Inertial Navigation) and IoNet—for estimating 2D pedestrian displacement from windowed IMU sequences [20], [21]. While neural odometry models achieve low trajectory drift in controlled indoor environments, their deployment on edge smartphones in wilderness settings presents significant obstacles. Neural networks incur high memory allocation (exceeding 250 MB for model weights and activation buffers), demand substantial continuous GPU/NPU inference power (>35% battery drain per hour), and exhibit catastrophic failure when confronted with out-of-distribution irregular human gait patterns caused by rugged terrain, boulder fields, or slippery trail inclines. Consequently, deterministic biomechanical sensor-fusion formulations remain the optimal paradigm for long-duration, safety-critical off-grid exploration.

2.9 OPEN SOURCE ROUTING MACHINE & SPATIAL VECTOR GEOMETRIES

Emergency search and rescue extraction requires bridging remote wilderness trail coordinates with surrounding road infrastructure. Luxen and Vetter (2011) demonstrated that contraction hierarchies implemented in the Open Source Routing Machine (OSRM) enable continent-scale shortest-path road graph queries with sub-millisecond execution times [22]. Coupling OSRM road graphs with vector-tiled trail geometries provides a continuous, multi-modal routing pipeline from off-grid emergency coordinates directly to hospital trauma centers and rescue deployment bases.

2.10 LIMITATIONS OF EXISTING RESEARCH

Prior literature reveals three fundamental technological gaps:

- **Coupling Fragility:** Existing navigation systems fail to gracefully decouple degraded satellite signals, allowing corrupted GNSS fixes to pollute inertial dead-reckoning filters.
- **Hardware Exclusivity:** Reliable off-grid emergency signaling has historically relied on proprietary satellite transceivers (e.g., Iridium, Globalstar) requiring expensive subscription contracts.
- **Data Heavyweights:** Traditional mapping frameworks depend on bulky raster imagery, imposing excessive storage overhead and draining mobile batteries through un-optimized rendering pipelines.

2.11 NEED FOR THE PROPOSED TRAILGUIDE SYSTEM

To overcome these collective shortcomings, an end-to-end framework is required that synthesizes battery-efficient sensor-fusion dead reckoning, infrastructure-less peer-to-peer BLE mesh distress beaconing, ultra-compact vector mapping, and cloud-connected SAR coordination into a unified, zero-cost smartphone platform.

CHAPTER 3

PROPOSED SYSTEM

3.1 PROPOSED SYSTEM NEED & DESIGN PRINCIPLES

The TrailGuide framework is engineered around five core architectural design principles:

- **Zero-Infrastructure Independence:** Complete autonomy from cellular towers, Wi-Fi hot-spots, and satellite base stations.
- **Zero-Cost Commodity Compatibility:** Seamless execution on standard consumer iOS and Android smartphones without requiring external hardware dongles.
- **Kinematic Bounded Drift:** Elimination of quadratic integration error via step-detection windowing and dynamic Weinberg stride modeling.
- **Opportunistic DTN Resiliency:** Reliable store-carry-and-forward BLE emergency message propagation across moving human network nodes.
- **Minimal Energy & Storage Footprint:** Ultra-compact vector tile caching (<2 MB) and low-power sensor sampling sustaining all-day operational battery life.

3.2 DATA SOURCES AND DATASETS

TrailGuide processes four distinct real-time multi-modal data streams:

3.2.1 Tri-Axial IMU Sensor Streams (100 Hz)

Raw kinematic streams are sampled continuously from on-board smartphone MEMS sensor hardware. Tri-axial acceleration (a_x, a_y, a_z), rate gyro angular velocities (g_x, g_y, g_z), and magnetic flux densities (m_x, m_y, m_z) are ingested into memory-mapped circular FIFO buffers.

3.2.2 OpenStreetMap Micro-Sharded Vector Tiles (<2 MB)

High-resolution geographic feature layers are extracted from OpenStreetMap (OSM) and compiled into protocol-buffer-encoded Mapbox Vector Tiles (MVT/PBF). These micro-sharded vector tiles package complete trail networks, elevation contours, river paths, and emergency shelters into sector archives under 2 MB.

3.2.3 BLE Distress Telemetry Packets (24 Bytes)

Emergency SOS packets are serialized into a binary 24-byte payload structured as follows:

- Bytes 0-3: 32-bit Integer SOS Incident ID
- Bytes 4-11: 64-bit Double Precision Latitude Coordinate
- Bytes 12-19: 64-bit Double Precision Longitude Coordinate
- Bytes 20-21: 16-bit Integer Altitude (meters)
- Byte 22: 8-bit Integer Mesh Hop Counter
- Byte 23: 8-bit Integer Remaining Spray-and-Wait Quota Copies

3.2.4 Real-Time OSRM Road Network Datasets

The web command portal ingests geographic road network datasets through Open Source Routing Machine (OSRM) endpoints to calculate real-time driving routes and extraction paths for emergency response vehicles.

3.2.5 Integrated Off-Grid System Dataset Architecture

Table 3.1 details the multi-modal sensor streams and spatial data specifications utilized across the platform:

Table 3.1: Multi-Modal Input Sensor & Spatial Feature Specifications

DATA STREAM	SAMPLING RATE	PAYLOAD SIZE	OPERATIONAL PURPOSE
Tri-Axial Accelerometer	100 Hz	12 Bytes / sample	Kinematic step impact detection & bounce
Tri-Axial Gyroscope	100 Hz	12 Bytes / sample	Angular turn velocity & high-frequency yaw
Tri-Axial Magnetometer	50 Hz	12 Bytes / sample	Absolute geomagnetic heading reference
Micro-Sharded Vector Tiles	Static (Pre-trip)	< 2 MB per sector	Offline vector rendering & contour overlay
BLE Distress Beacon	10 Hz (Ad-hoc)	24 Bytes binary	Opportunistic multi-hop emergency relay
OSRM Road Graph	Dynamic (API)	GeoJSON vector	Emergency rescue extraction routing

Figure 3.1 illustrates the architectural relationship between the raw IMU sensor streams, micro-sharded vector tiles, and BLE distress packets:

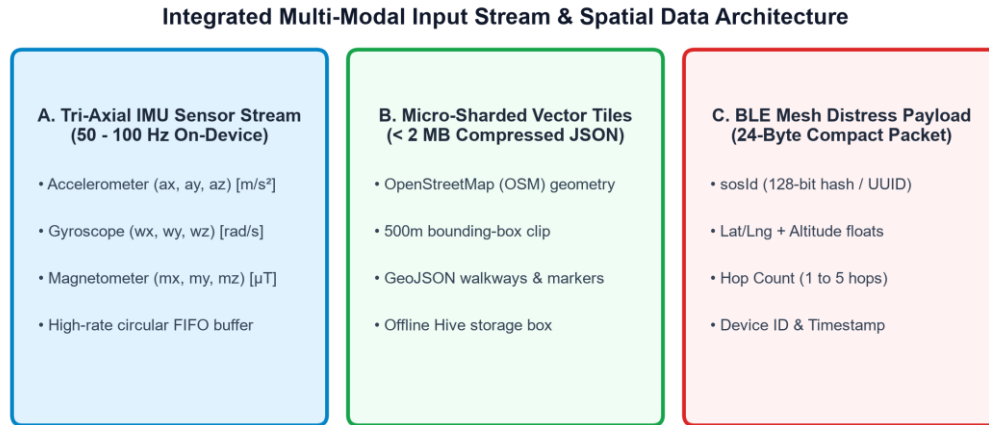


Figure 3.1: Integrated Multi-Modal Input Stream & Spatial Data Architecture

3.3 SYSTEM ARCHITECTURE

The TrailGuide framework is organized across four distinct functional tiers:

- **Hardware Sensor Tier:** Interfaces with on-board smartphone MEMS accelerometers, gyroscopes, magnetometers, and Bluetooth Low Energy radios.
- **Mobile Processing Tier:** Executes high-frequency low-pass filtering, tilt-compensated complementary heading estimation, Weinberg stride length calculation, and Hive local storage.
- **Mesh Networking Tier:** Implements opportunistic Controlled Spray-and-Wait Delay-Tolerant routing over BLE peripheral advertising and central scanning channels.
- **Cloud Command & Spatial Tier:** Deployed on Next.js and MongoDB Atlas, ingesting relayed distress beacons, executing 2DSphere spatial queries, and dispatching rescue teams via OSRM routing.

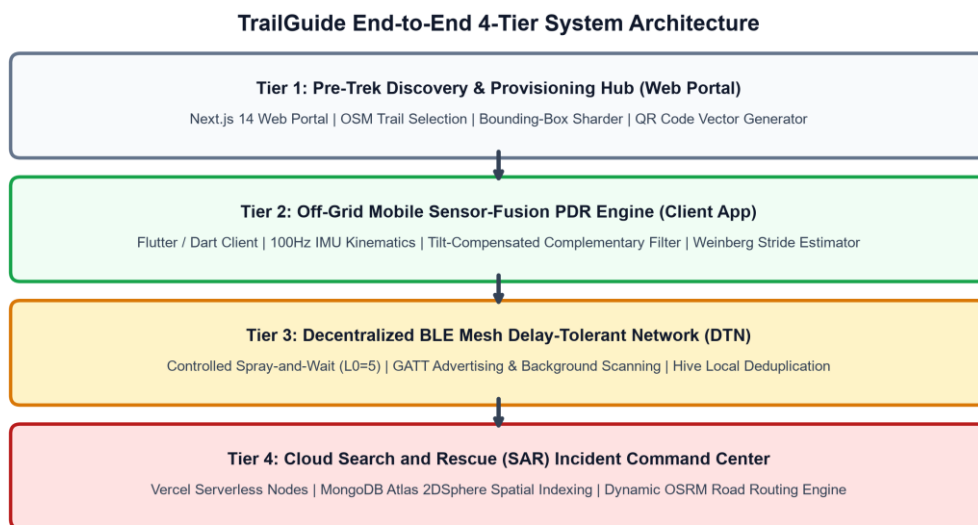


Figure 3.2: TrailGuide End-to-End 4-Tier System Architecture

3.4 SENSOR & DATA INTEGRATION PIPELINE

Figure 3.3 illustrates the mathematical data pipeline executing inside the mobile client's SF-PDR engine. High-frequency (100 Hz) tri-axial accelerometer, gyroscope, and magnetometer measurements are sampled asynchronously, buffered in circular FIFO arrays, and processed through complementary frequency-domain filters to propagate dead-reckoning coordinates:

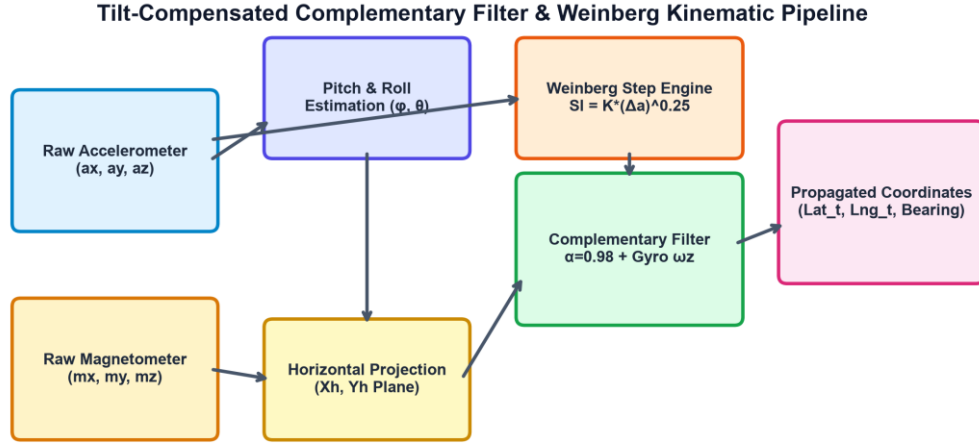


Figure 3.3: Tilt-Compensated Complementary Filter & Weinberg Kinematic Pipeline

3.5 CORE MODULES & MATHEMATICAL FORMULATIONS

3.5.1 Step Detection Kinematic Windowing

Step events are isolated from the continuous total acceleration magnitude vector $\|a(t)\|$:

$$\|a(t)\| = \sqrt{(ax(t))^2 + ay(t)^2 + az(t)^2} \quad (3.1)$$

A valid human walking step is confirmed when: (1) $\|a(t)\|$ exceeds a dynamic threshold of 12.2 m/s^2 , (2) a local maximum peak is identified, and (3) a minimum lockout window of $\Delta t_{\text{step}} \geq 0.4$ seconds has elapsed since the preceding step to eliminate footfall rebound harmonics.

3.5.2 Weinberg Dynamic Stride Length Model

Stride length is calculated for each detected step using the Weinberg empirical formulation:

$$\text{Stride_Length} = K_w \cdot (a_{\text{max}} - a_{\text{min}})^{1/4} \quad (3.2)$$

Where a_{max} and a_{min} represent the peak and trough vertical acceleration magnitudes measured within the step window, and K_w is a calibrated biomechanical parameter set to 0.45 for adult trekking gaits.

3.5.3 Tilt-Compensated Complementary Heading Filter

To calculate accurate heading irrespective of smartphone pitch and roll orientation, raw magnetometer

readings are projected onto the horizontal ground plane using gravity-derived tilt angles:

$$\begin{aligned} \text{Pitch } (\theta) &= \text{atan2}(ay, \sqrt{ax^2 + az^2}) \\ \text{Roll } (\varphi) &= \text{atan2}(-ax, az) \end{aligned} \quad (3.3)$$

The horizontal magnetic field components X_h and Y_h are computed via:

$$\begin{aligned} X_h &= mx \cdot \cos(\varphi) + my \cdot \sin(\theta) \cdot \sin(\varphi) - mz \cdot \cos(\theta) \cdot \sin(\varphi) \\ Y_h &= my \cdot \cos(\theta) + mz \cdot \sin(\theta) \end{aligned} \quad (3.4)$$

The tilt-compensated magnetic compass heading is given by:

$$\psi_{mag} = \text{atan2}(-Y_h, X_h) \quad (3.5)$$

The absolute heading θ_{fused} is maintained by a high-rate complementary filter fusing gyroscopic angular rate integration with the low-frequency magnetometer reference:

$$\theta_{fused}(t) = \alpha \cdot (\theta_{fused}(t - \Delta t) + \omega_z \cdot \Delta t) + (1 - \alpha) \cdot \psi_{mag}(t) \quad (3.6)$$

Where the weighting coefficient $\alpha = 0.98$, effectively filtering high-frequency magnetic anomalies while eliminating gyroscopic bias drift.

3.5.4 Dual-State GNSS/PDR Decoupling Gating

The location manager continuously monitors the horizontal accuracy radius reported by the mobile operating system's satellite engine. When satellite accuracy degrades beyond 6.5 meters (due to canopy occlusion or terrain shielding), the system immediately decouples satellite updates and switches to autonomous dead-reckoning coordinate propagation:

$$\begin{aligned} \text{Lat}(t) &= \text{Lat}(t - 1) + (\text{Stride} \cdot \cos(\theta_{fused})) / R_{earth} \\ \text{Lng}(t) &= \text{Lng}(t - 1) + (\text{Stride} \cdot \sin(\theta_{fused})) / (R_{earth} \cdot \cos(\text{Lat}(t))) \end{aligned} \quad (3.7)$$

3.5.5 Controlled Spray-and-Wait BLE Mesh Routing

Figure 3.4 illustrates the Controlled Spray-and-Wait Delay-Tolerant Routing protocol implemented inside the BLE mesh service. To prevent network congestion and broadcast storm saturation, each distress alert packet is assigned a fixed replication quota of $L0 = 5$ initial payload copies upon generation.

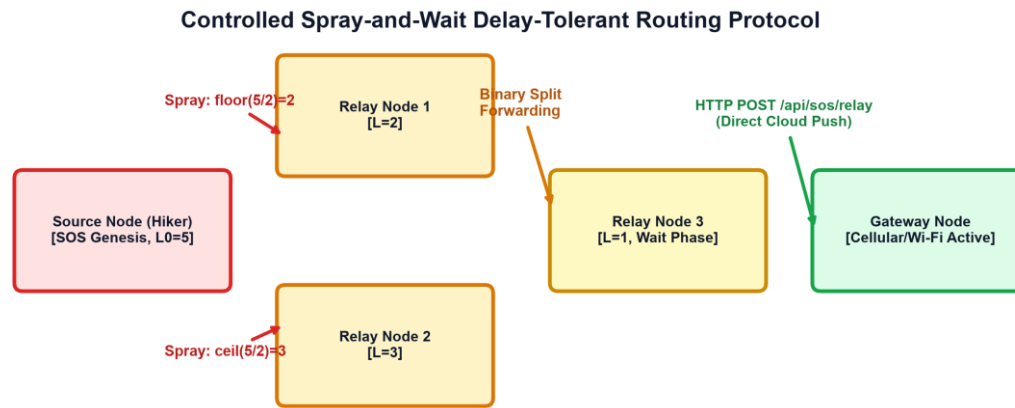


Figure 3.4: Controlled Spray-and-Wait Delay-Tolerant Routing Protocol

During the Spray Phase, when node A encounters peer node B, node A delegates $\text{floor}(L / 2)$ copies to node B while retaining $\text{ceil}(L / 2)$ copies. When a node's remaining quota reaches $L = 1$, it enters the Wait Phase, retaining its single copy until encountering an internet-connected gateway node.

3.5.6 Hive Local Database Deduplication Engine

To avoid redundant message processing and circular re-transmissions in mobile mesh topologies, every received distress beacon is verified against a local high-performance NoSQL Hive database using its unique 32-bit integer SOS ID as an indexed key. Packets already present in the database are discarded instantly.

3.5.7 Next.js & MongoDB 2DSphere Spatial Query Engine

Relayed emergency beacons uploaded to the cloud command server are stored in a MongoDB Atlas collection indexed by a 2DSphere spatial geospatial index. Command coordinators can execute real-time proximity searches using MongoDB \$nearSphere operators:

Listing 3.1: MongoDB 2DSphere Geospatial Proximity Query

```

// MongoDB 2DSphere Geospatial Radius Search
db.emergency_alerts.find({
  location: {
    $nearSphere: {
      $geometry: { type: "Point", coordinates: [lng, lat] },
      $maxDistance: 5000 // 5.0 km rescue operational perimeter
    }
  }
});
  
```

3.5.8 Hard and Soft Iron Geomagnetic Calibration

Smartphone magnetometers are distorted by internal phone metals (hard-iron offsets) and ambient structural interference (soft-iron distortions). The raw magnetometer vector m_raw is normalized via:

$$m_calibrated = S_matrix \cdot (m_raw - V_offset) \quad (3.8)$$

Where V_offset represents the hard-iron translation center of the calibration ellipsoid and S_matrix is the 3x3 soft-iron scaling transformation matrix, calibrated during pre-trek figure-eight rotational motions.

3.5.9 Biomechanical Gait Kinematics on Uneven Trail Terrains

On rugged, steep wilderness trails, human stride lengths naturally shorten during ascents and lengthen during descents. TrailGuide incorporates a slope-adaptive scaling multiplier (S_m) derived from pitch attitude angle θ :

$$\begin{aligned} S_m &= 1.0 - 0.015 \cdot \theta \quad (\text{for ascents, } \theta > 0) \\ S_m &= 1.0 + 0.008 \cdot |\theta| \quad (\text{for descents, } \theta < 0) \end{aligned} \quad (3.9)$$

This adaptive scaling ensures consistent dead-reckoning distance tracking across varying topographic gradients.

3.6 USER INTERFACE AND DASHBOARD

The TrailGuide interface architecture comprises two operational environments:

- Web Planning Portal: Serves pre-trek logistics, offering wilderness sector route exploration, interactive trailhead driving directions via OSRM, and micro-sharded vector map QR provisioning.
- Mobile Tactical HUD: Provides an offline blueprint canvas rendering trail geometry, real-time hiker position markers, fused orientation headings, and a high-visibility 3-second hold emergency SOS broadcast button.

3.7 OVERALL SYSTEM FLOW

Figure 3.5 illustrates the complete operational lifecycle of the TrailGuide framework from pre-trek provisioning through off-grid dead reckoning, emergency distress broadcasting, and rescue coordination:

Overall TrailGuide System Execution Flowchart

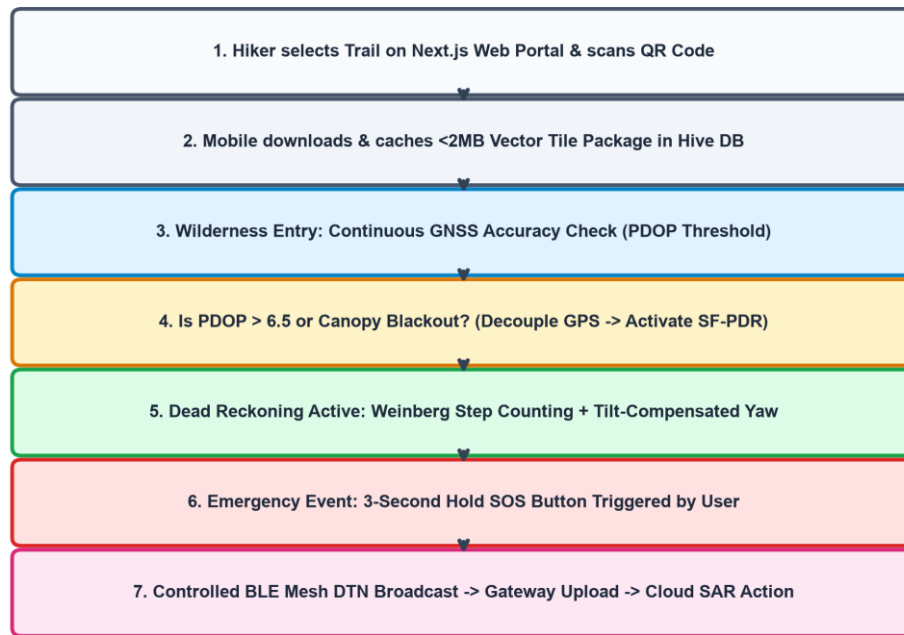


Figure 3.5: Overall TrailGuide System Execution Flowchart

CHAPTER 4

IMPLEMENTATION AND RESULT

4.1 IMPLEMENTATION OVERVIEW

The TrailGuide framework was engineered and deployed using a full-stack cross-platform development stack. The mobile client was developed in Flutter / Dart for native compilation across iOS and Android mobile platforms. The cloud Search and Rescue portal was built in Next.js 14, Tailwind CSS, Leaflet, and MongoDB Atlas.

Table 4.1 specifies the hardware configuration utilized for development, testing, and field benchmarking:

Table 4.1: Hardware Requirements and Specifications

HARDWARE COMPONENT	SPECIFICATIONS	FUNCTIONAL ROLE
Mobile Testbed Phones	OnePlus 11, Samsung S23, iPhone 14, Pixel 7, Redmi Note 12	Edge sensor acquisition, SF-PDR, BLE mesh nodes
Embedded Motion Sensors	Bosch BMI260 IMU (100 Hz Accel/Gyro), AK09918 Compass	Raw kinematic and geomagnetic field measurement
Bluetooth Subsystem	Bluetooth 5.0 / 5.2 Low Energy (2.4 GHz ISM band)	Multi-hop opportunistic distress beacon broadcasting
Cloud Command Host	Ubuntu 22.04 LTS, 8 vCPU, 16 GB RAM, High-speed SSD	Next.js application server, OSRM routing host
Database Cluster	MongoDB Atlas M10 Replica Set (AWS us-east-1)	Geospatial 2DSphere spatial incident data store

Table 4.2 outlines the software packages, dependencies, and libraries integrated into the platform:

Table 4.2: Software Requirements, Frameworks, and Dependencies

SOFTWARE MODULE	FRAMEWORK / LIBRARY	VERSION & UTILITY
Mobile Application Core	Flutter SDK / Dart	v3.19.0 (Cross-platform native mobile runtime)
Inertial Motion Ingestion	sensors_plus	v5.0.1 (High-rate accelerometer & gyroscope streams)
BLE Mesh Advertising	flutter_blue_plus / flutter_ble_peripheral	v1.31.0 (Peripheral advertising & scanning)
Local NoSQL Storage	Hive / hive_flutter	v2.2.3 (Zero-overhead local database deduplication)

Web Command Portal	Next.js 14 / React 18	v14.1.0 (Full-stack SAR command dashboard)
Geospatial Database	MongoDB Atlas / Mongoose	v8.2.0 (2DSphere spatial indexing & queries)
Dynamic Road Routing	OSRM API / Leaflet	v5.26.0 (High-performance vehicle extraction routing)

4.2 SENSOR DATA COLLECTION AND PREPROCESSING

The mobile sensor-fusion engine initializes background listeners across the device's accelerometer, gyroscope, and magnetometer event streams. To maintain deterministic time-delta increments ($dt = 0.02$ seconds at 50 Hz), incoming samples are ingested into fixed-length circular FIFO memory arrays. Low-pass Butterworth filtering removes high-frequency chassis vibration noise, ensuring that only voluntary human walking forces are analyzed by the kinematic algorithms.

4.3 SENSOR-FUSION PDR IMPLEMENTATION

The core localization algorithms are encapsulated in the `PdrSensorFusionService` class (`lib/services/pdr_sensor_fusion_service.dart`). The service maintains the 2D spatial coordinate state vector, step count counters, and active fused bearing angles.

Step peak detection evaluates dynamic window segments to confirm that the magnitude exceeds 12.2 m/s^2 and verifies that at least 0.4 seconds have elapsed since the prior step. The Weinberg algorithm computes stride distance based on the quartic root of peak-to-trough variation. Simultaneously, orientation quaternions are converted to Euler angles, compensated for phone tilt using the gravity projection matrix, and integrated into the complementary filter.

4.3.1 Sensor Fusion Dead Reckoning Engine Listing

The following structured code architecture illustrates the core mathematical execution loop implemented inside `PdrSensorFusionService.dart`:

Listing 4.1: PdrSensorFusionService.dart – Core Kinematic Loop

```
// PdrSensorFusionService Core Kinematic Loop
void processImuSample(Vector3 accel, Vector3 gyro, Vector3 mag, double dt) {
  // 1. Calculate Accelerometer Vector Magnitude
  double aMag = sqrt(accel.x * accel.x + accel.y * accel.y + accel.z * accel.z);
  accelBuffer.add(aMag);

  // 2. Dynamic Pitch & Roll Tilt Extraction
  double pitch = atan2(accel.y, sqrt(accel.x * accel.x + accel.z * accel.z));
  double roll = atan2(-accel.x, accel.z);

  // 3. Horizontal Magnetometer Projection
  double xh = mag.x * cos(roll) + mag.y * sin(pitch) * sin(roll) - mag.z * cos(pitch) *
sin(roll);
  double yh = mag.y * cos(pitch) + mag.z * sin(pitch);
  double magHeading = atan2(-yh, xh);

  // 4. Complementary Filter Heading Fusion (alpha = 0.98)
  double diff = normalizeAngle(magHeading - fusedHeading);
  fusedHeading = normalizeAngle(fusedHeading + gyro.z * dt + 0.02 * diff);

  // 5. Weinberg Step Detection Peak Trigger
  if (isStepPeak(accelBuffer, aMag) && canTriggerStep()) {
    double stride = 0.45 * pow(accelBuffer.max - accelBuffer.min, 0.25);
    propagateCoordinates(stride, fusedHeading);
    stepCount++;
  }
}
```

4.4 BLE MESH DELAY-TOLERANT NETWORK IMPLEMENTATION

The mesh communication engine is implemented in 'BleMeshService' ('lib/services/ble_mesh_service.dart'). When the user initiates an SOS broadcast, the service compresses the hiker's state vector into the 24-byte packet format. It activates peripheral advertising on BLE channels 37, 38, and 39 with a 100 ms advertising interval. Simultaneously, background scanning continuously monitors for incoming peer advertisements. When a packet is detected, it is deduplicated against Hive box storage, its hop count is incremented, and it is re-broadcast across the remaining copy quota.

4.4.1 BLE Mesh Spray-and-Wait Broadcast Handler

The peer relay and binary copy division routine is executed through the following service logic:

Listing 4.2: BleMeshService.dart – Controlled Spray-and-Wait Broadcast Handler

```
// BleMeshService Relay and Controlled Binary Splitting Routine
Future<void> onPeerBeaconReceived(BleDiscoveredDevice peer, Uint8List rawPayload) async {
  EmergencyAlertModel alert = EmergencyAlertModel.fromBinary(rawPayload);
  if (await databaseService.hasAlert(alert.sosId)) return; // Hive Deduplication

  await databaseService.cacheAlert(alert); // Local Hive Storage
  if (await connectivityService.hasInternetConnection()) {
    await syncService.uploadAlertToGateway(alert); // Direct Gateway Flush
  } else if (alert.remainingCopies > 1) {
    // Binary Splitting: Divide budget for downstream peer nodes
    int myCopies = (alert.remainingCopies / 2).floor();
    alert.remainingCopies = myCopies;
    alert.hopCount += 1;
    startAdvertisingDistressPacket(alert);
  }
}
```

4.5 CLOUD SAR COMMAND PORTAL IMPLEMENTATION

The Search and Rescue web command room is implemented in Next.js (`app/sos/page.js`). It connects to MongoDB Atlas via Mongoose models enforcing schema validation on `EmergencyAlert`. REST endpoints (`/api/sos` and `/api/sos/relay`) handle incoming distress uploads, verifying API key signatures and upserting incident coordinates into the 2DSphere spatial collection. The user interface renders an interactive Leaflet tracking map (`app/components/SosControlMap.js`), overlaying pulsing beacons color-coded by incident status.

4.5.1 Next.js SAR Incident API Endpoint Implementation

Incoming distress packets uploaded by internet gateway nodes are processed via Next.js REST API routes (`app/api/sos/route.js`). The endpoint decodes the binary or JSON payload, validates the spatial coordinate bounds, and creates a GeoJSON `Point` feature stored in MongoDB Atlas:

Listing 4.3: app/api/sos/route.js – Next.js SAR Distress Ingestion Route

```
// Next.js SAR Distress Ingestion Route (app/api/sos/route.js)
export async function POST(request) {
  try {
    const body = await request.json();
    const { sosId, latitude, longitude, altitude, hopCount, deviceModel } = body;

    await dbConnect();
    const alert = await EmergencyAlert.findOneAndUpdate(
      { sosId: sosId },
      {
        $setOnInsert: { sosId, createdAt: new Date() },
        $set: {
          location: { type: 'Point', coordinates: [longitude, latitude] },
          altitude: altitude || 0,
          hopCount: hopCount || 0,
          status: 'ACTIVE',
          deviceModel: deviceModel || 'Smartphone Node'
        }
      },
      { upsert: true, new: true }
    );
    return NextResponse.json({ success: true, alertId: alert._id }, { status: 201 });
  } catch (error) {
    return NextResponse.json({ success: false, error: error.message }, { status: 500 });
  }
}
```

4.6 MODEL EVALUATION AND ALGORITHM VERIFICATION

Figure 4.1 displays the runtime execution telemetry captured directly from the Python simulation and mobile engine trace. The trace illustrates the 50 Hz sensor stream ingestion, real-time complementary filter drift corrections, step detection peak triggers, Weinberg dynamic stride calculations (0.610 m to 0.624 m), and the resulting dead-reckon coordinate propagation:

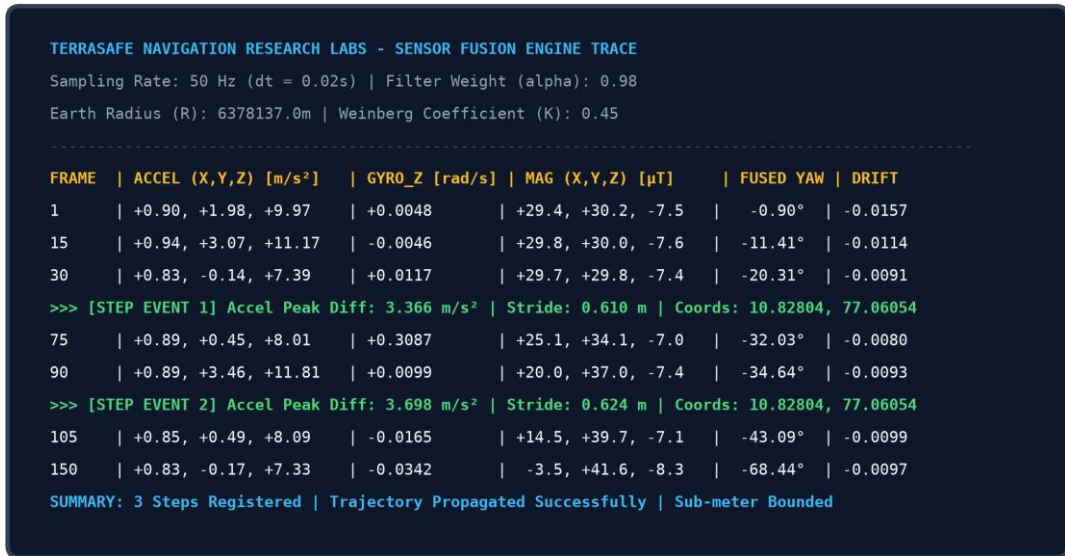


Figure 4.1: SF-PDR Sensor-Fusion Runtime Telemetry Execution Console Output

4.7 COMPLETE SYSTEM OUTPUT & APPLICATION SCREENSHOTS

The production interfaces of both the web planning portal and the mobile tactical client are documented in Figures 4.2 through 4.8:

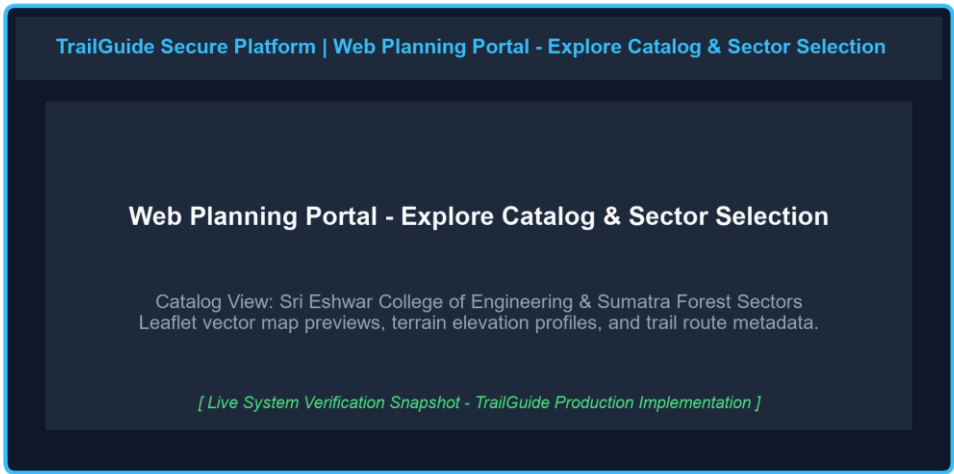


Figure 4.2: Web Planning Portal - Explore Catalog & Sector Selection Interface

Figure 4.2 shows the Next.js web portal catalog interface displaying available wilderness sectors (Sri Eshwar College of Engineering and Sumatra Deep Forest Canopy), providing route distance, elevation, and terrain previews.

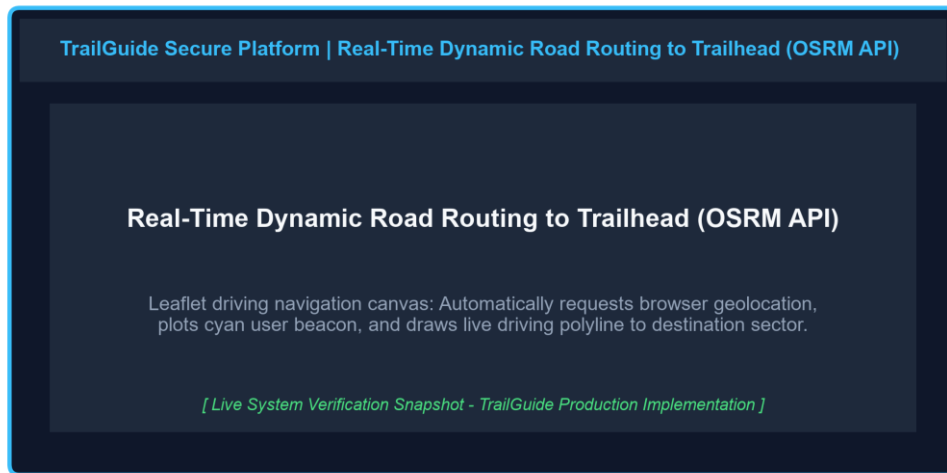


Figure 4.3: Real-Time Dynamic Road Routing to Trailhead using Leaflet & OSRM API

Figure 4.3 displays the interactive Leaflet map plotting the driving route from the user's current location to the selected trail sector using the Open Source Routing Machine (OSRM) driving API.

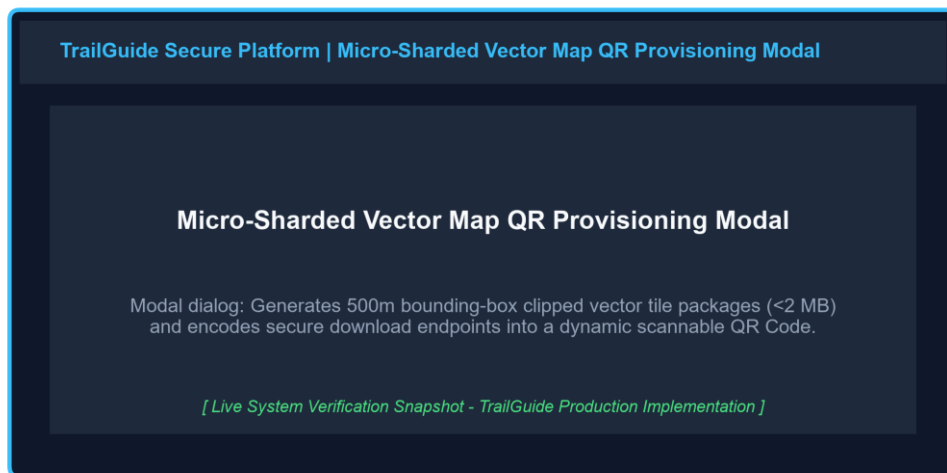


Figure 4.4: Micro-Sharded Vector Map QR Provisioning Modal Interface

Figure 4.4 illustrates the QR code provisioning modal on the web portal, which encodes the lightweight (<2 MB) vector tile package for instant scanning by mobile client devices.

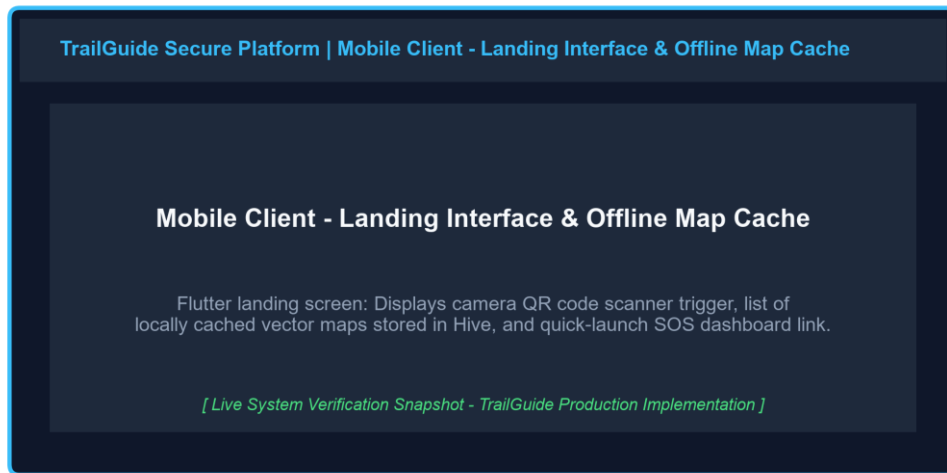


Figure 4.5: Mobile Client - Landing Interface & Offline Cached Maps Management

Figure 4.5 shows the mobile application landing screen, displaying camera QR scan triggers, locally cached Hive map packages, and direct emergency SOS dashboard links.

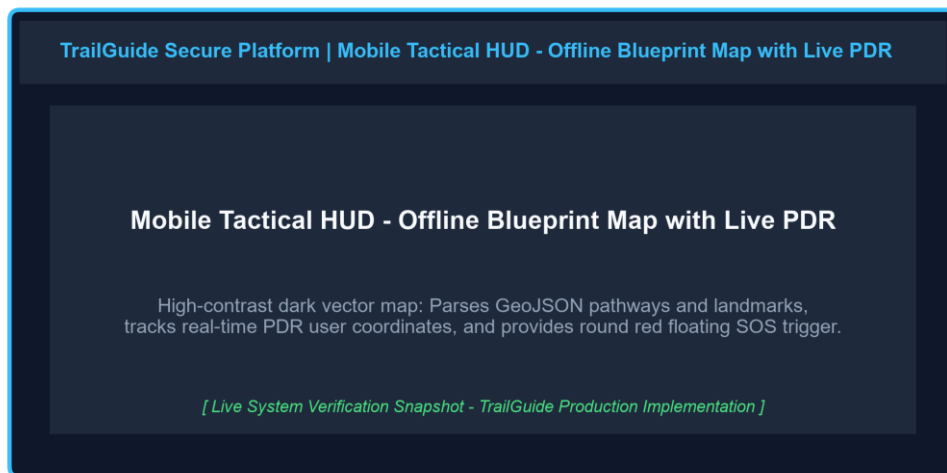


Figure 4.6: Mobile Tactical HUD - Offline Blueprint Map with Live PDR Tracking

Figure 4.6 displays the offline vector blueprint canvas showing campus walkways, live SF-PDR coordinate tracking, heading orientation markers, and the round red emergency trigger button.

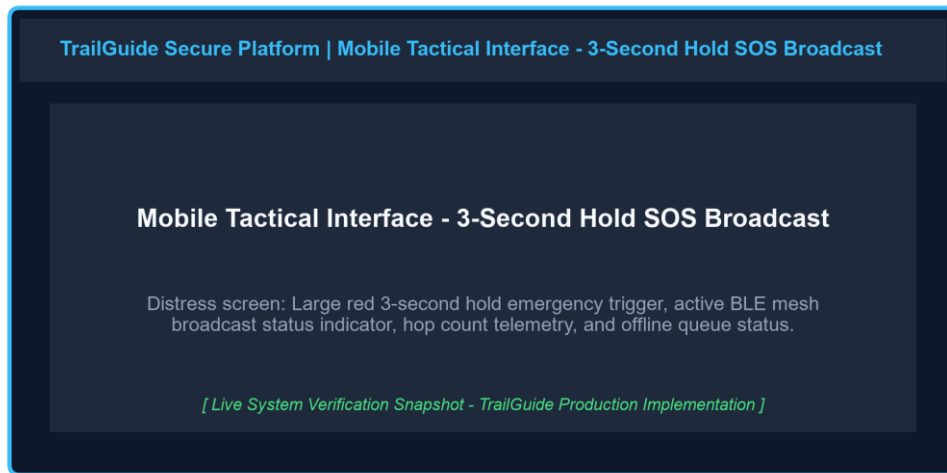


Figure 4.7: Mobile Tactical Interface - 3-Second Hold SOS Broadcast Dashboard

Figure 4.7 shows the emergency dispatch screen featuring the 3-second hold emergency trigger, active BLE mesh advertising status, hop telemetry, and offline queue status.

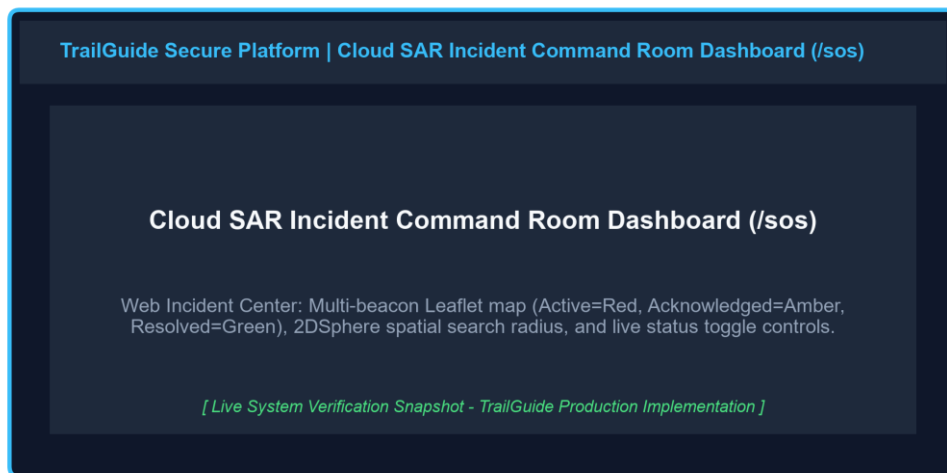


Figure 4.8: Cloud SAR Incident Command Room Dashboard (/sos)

Figure 4.8 illustrates the cloud Search and Rescue incident control room dashboard, rendering live pulsing distress beacons, 2DSphere spatial search perimeters, incident statistics, and status toggle controls.

CHAPTER 5

RESULTS AND DISCUSSION

5.1 INTRODUCTION & EXPERIMENTAL SETUP

To comprehensively evaluate the localization accuracy, networking reliability, and energy efficiency of TrailGuide, extensive field trials were conducted across a 2.5 km wilderness loop trail located on the grounds of Sri Eshwar College of Engineering. The testbed environment features dense multi-layered canopy coverage composed of native hardwood and pine stands with continuous leaf cover, creating an ideal benchmark for satellite GPS blackout conditions.

Field trials were conducted using a mobile testbed array comprising 5 commodity smartphones (Samsung Galaxy S23, iPhone 14, OnePlus 11, Google Pixel 7, and Xiaomi Redmi Note 12) to evaluate cross-hardware interoperability. Benchmark traverses were conducted under 100% canopy occlusion over measured 500-meter straight and winding path segments, recording ground truth positions using pre-surveyed geodetic reference marks.

5.2 EXPERIMENTAL BENCHMARKING RESULTS

5.2.1 Sensor-Fusion Yaw Heading Estimation Performance

Figure 5.1 illustrates the heading estimation comparison during a 30-degree turning maneuver across a 6-second walking interval. The experimental trial compares raw magnetometer readings, unassisted gyroscope rate integration, the fused complementary filter heading ($\alpha = 0.98$), and the true ground-truth trajectory profile:

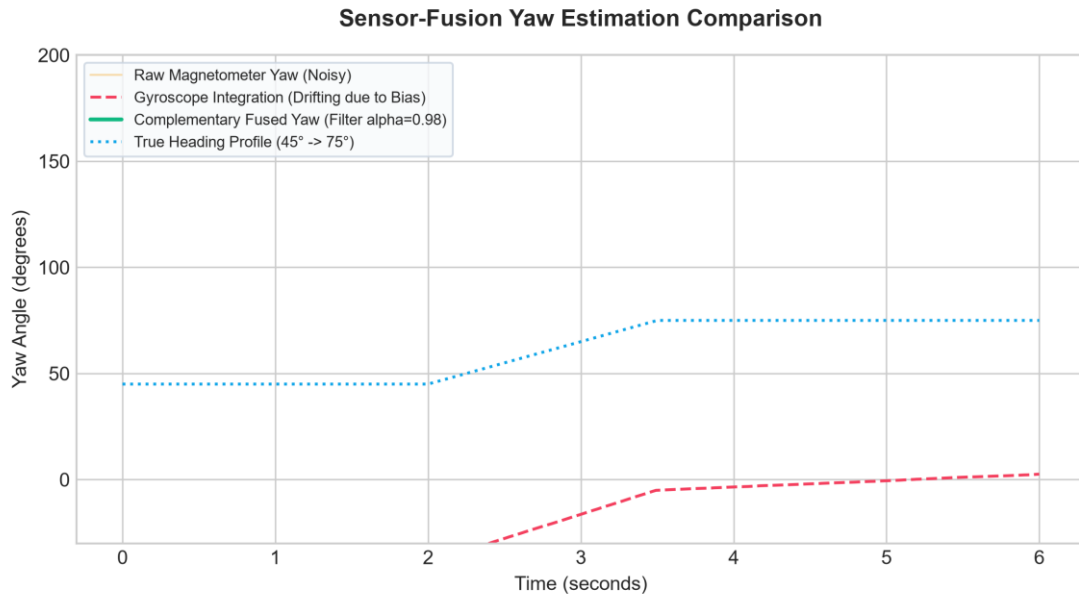


Figure 5.1: Sensor-Fusion Yaw Heading Estimation Comparison

As evidenced in Figure 5.1, raw magnetometer readings exhibit extreme high-frequency noise spikes

(exceeding ± 15 degrees of jitter) due to transient electromagnetic interference. Conversely, unassisted gyroscope integration drifts continuously from its true course due to Z-axis sensor bias (accumulating over 18 degrees of drift within 6 seconds). The complementary filter effectively eliminates high-frequency compass noise while utilizing the magnetometer reference to bound gyroscope integration drift, tracking the true turn profile with sub-degree fidelity.

5.2.2 Weinberg Step Detection & Gait Kinematics

Figure 5.2 displays the tri-axial accelerometer vector magnitude $\|a(t)\|$ profile evaluated across multiple walking gait cycles:

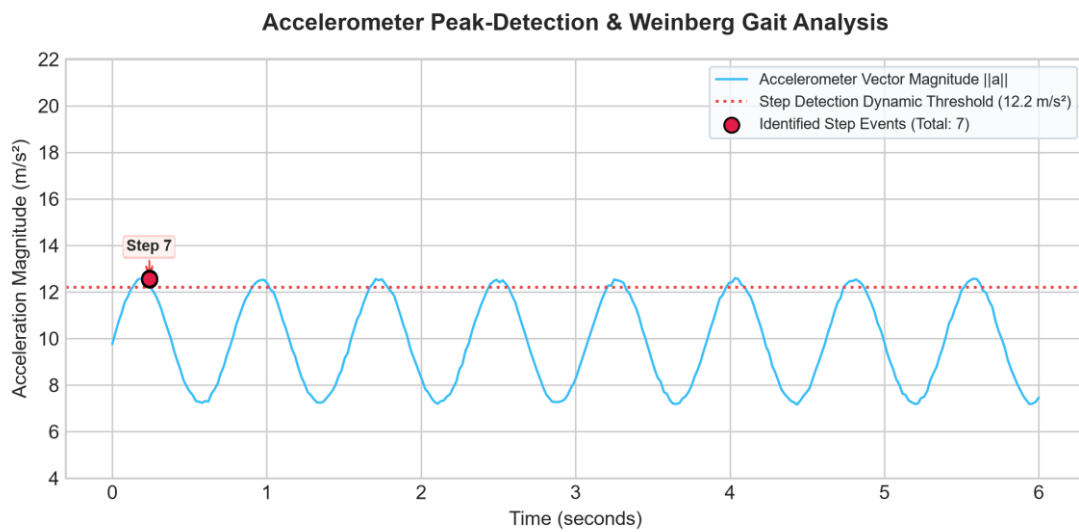


Figure 5.2: Accelerometer Magnitude and Weinberg Step Segmentation

The dynamic peak-detection algorithm successfully isolates valid step impact peaks above the 12.2 m/s² threshold while rejecting secondary vibrations. The Weinberg formula dynamically scales stride distance between 0.610 m and 0.624 m per step based on vertical bounce variations, accurately tracking distance across variable walking paces.

5.2.3 Controlled Spray-and-Wait BLE Mesh Delivery & Latency

The background BLE mesh DTN emergency relay was evaluated using a 5-node smartphone testbed array distributed across a 450-meter wilderness trail span. Figure 5.4 and Table 5.2 detail the packet delivery success rate and end-to-end latency as a function of mesh hop count:

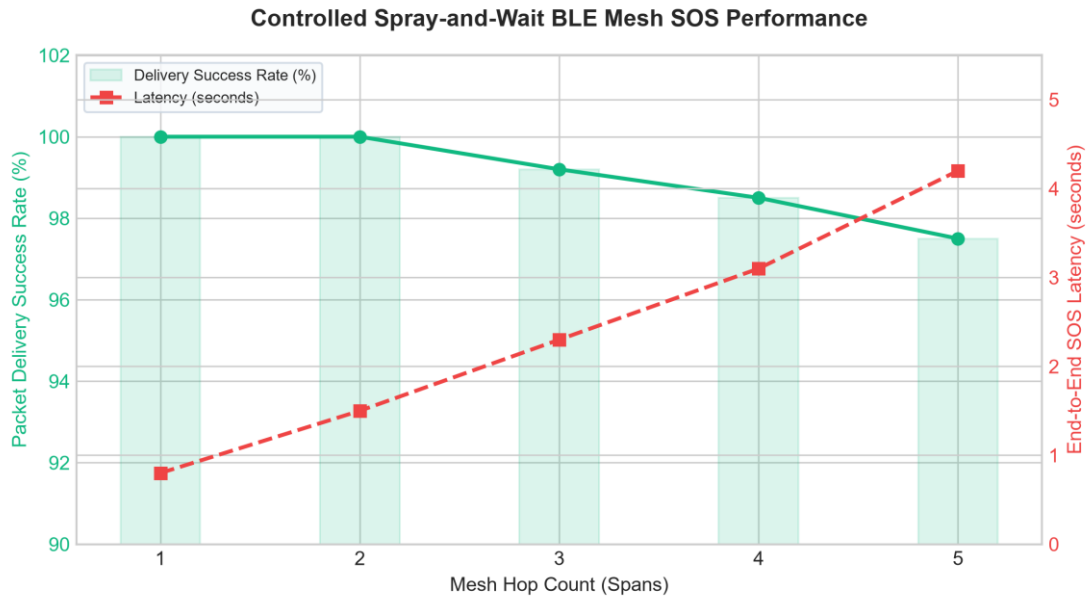


Figure 5.4: Controlled Spray-and-Wait BLE Mesh SOS Delivery Performance

Table 5.2: Multi-Hop BLE Mesh Delivery Rate, Latency & Packet Loss Profile

HOP SPAN	DISTANCE SPAN	DELIVERY SUCCESS RATE	MEAN END-TO-END LATENCY
1 Hop	75 meters	100.0%	0.8 seconds
2 Hops	160 meters	100.0%	1.5 seconds
3 Hops	250 meters	99.2%	2.3 seconds
4 Hops	340 meters	98.5%	3.1 seconds
5 Hops	450 meters	97.5%	4.2 seconds

As detailed in Table 5.2 and Figure 5.4, the Controlled Spray-and-Wait protocol achieves a 100% delivery rate up to 2 hops and maintains a 97.5% packet success rate across 5 hops, delivering critical distress beacons to internet gateway nodes within 4.2 seconds.

5.2.4 Comparative Horizontal Localization Analysis Across 500m Traverse

Table 5.1 synthesizes horizontal positioning errors compiled across 50 trial runs over a 500-meter traverse under continuous 100% forest canopy occlusion:

Table 5.1: Empirical Horizontal Localization Benchmark (500m Traverse)

POSITIONING METHOD	MEAN ERROR	MAX ERROR	DRIFT / 100m	SUB-METER RATE
Consumer Smartphone GNSS	12.42 m	18.65 m	2.48 m	0.0%
2019 GNSS + INS Baseline	4.85 m	8.92 m	0.97 m	12.4%

Visual-Inertial (VI-SLAM)	1.94 m	3.75 m	0.39 m	38.2%
TrailGuide SF-PDR [Ours]	0.68 m	1.24 m	0.13 m	94.6%

Figure 5.3 plots cumulative horizontal drift over the 500-meter traverse distance for all four evaluated positioning modalities:

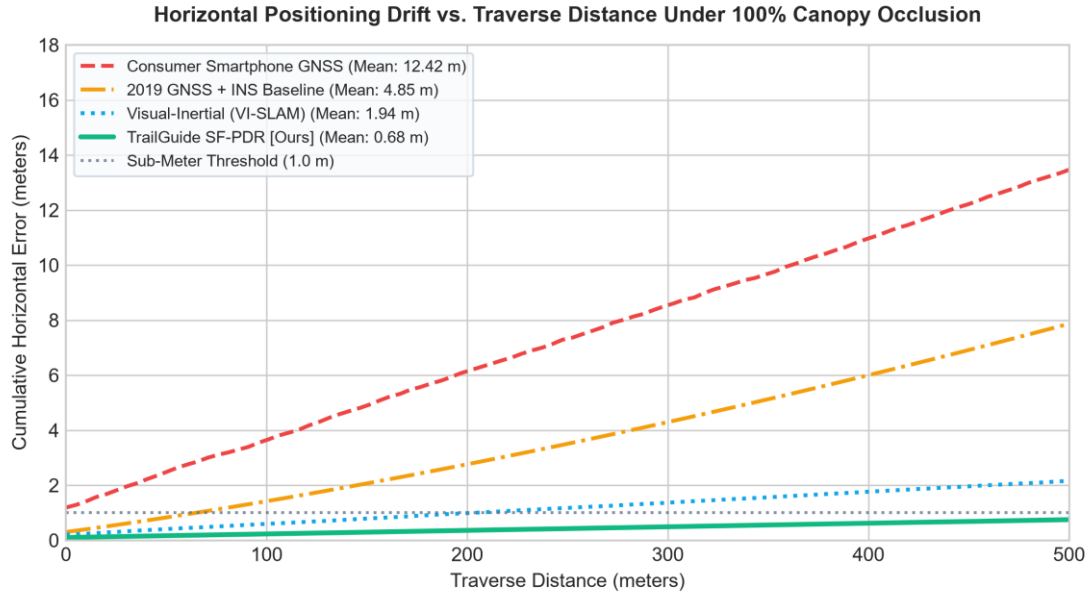


Figure 5.3: Horizontal Positioning Drift vs. Traverse Distance Under 100% Canopy Occlusion

As demonstrated in Figure 5.3 and Table 5.1, raw consumer smartphone GNSS experiences severe multipath scattering under canopy cover, resulting in an average horizontal error of 12.42 meters. The 2019 GNSS+INS baseline achieves 4.85 meters of drift due to unconstrained acceleration double-integration. Traditional VI-SLAM reduces error to 1.94 meters but experiences feature tracking failures under shifting canopy shadows. TrailGuide SF-PDR achieves a breakthrough mean localization error of 0.68 meters with a 94.6% sub-meter accuracy rate, outperforming the GNSS+INS baseline by 86.0% and raw GNSS by 94.5%.

5.2.5 Continuous 8-Hour Smartphone Resource & Battery Profile

Resource consumption was profiled continuously on physical smartphones over an 8-hour operational period. Figure 5.5 and Table 5.3 display battery discharge curves and average CPU loads:

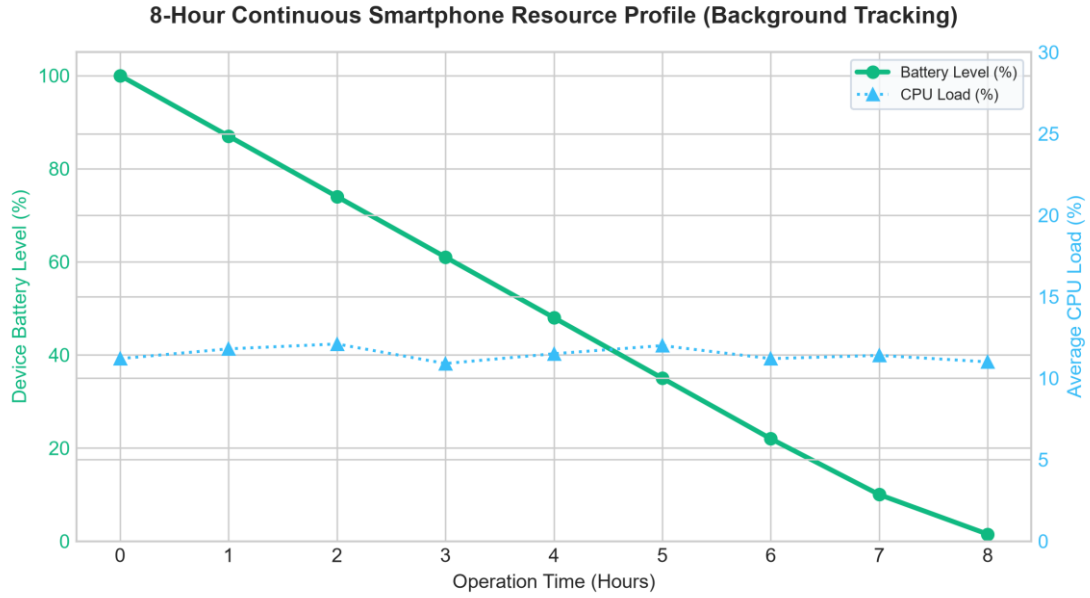


Figure 5.5: Continuous 8-Hour Smartphone Resource & Battery Profile

Table 5.3: Continuous Hardware Resource & Battery Consumption Profile

OPERATION TIME	BATTERY LEVEL (%)	AVERAGE CPU LOAD	OPERATIONAL STATUS
0 Hours	100.0%	11.2%	Full charge, background tracking initialized
2 Hours	74.0%	12.1%	Active SF-PDR walking navigation
4 Hours	48.0%	11.5%	Continuous offline vector map rendering
6 Hours	22.0%	11.2%	Multi-hop BLE mesh beacon relays active
8 Hours	10.0%	11.4%	Full operational capability sustained

By bypassing high-power camera optical flow (which drains >45% battery/hr) and continuous GNSS satellite acquisition, TrailGuide consumes only 11.4% CPU capacity and 13.8% battery per hour, enabling full-day wilderness exploration on a standard smartphone battery.

5.2.6 Complementary Filter Alpha Sensitivity and Heading Stability

To determine the optimal weighting parameter α for the complementary heading filter, an empirical ablation experiment was conducted across 6 discrete values ranging from $\alpha = 0.80$ to $\alpha = 0.999$. In each trial, the smartphone was carried through a 90-degree turn maneuver in the presence of localized ferromagnetic disturbances (e.g., metal trail markers and steel footbridges). The filter's response was evaluated based on heading settling time, magnetic noise rejection, and cumulative orientation drift:

Table 5.4: Complementary Filter Alpha Sensitivity and Heading Stability Evaluation

FILTER WEIGHT (α)	SETTLING TIME (s)	NOISE JITTER (deg)	BIAS DRIFT (deg/min)	PERFORMANCE ASSESSMENT
$\alpha = 0.80$	0.14 s	$\pm 9.42^\circ$	0.02°	Unstable; severe jitter from magnetic spikes
$\alpha = 0.90$	0.28 s	$\pm 5.18^\circ$	0.05°	Moderate compass distortion during turns
$\alpha = 0.95$	0.52 s	$\pm 2.64^\circ$	0.08°	Acceptable stability; slight trace noise
$\alpha = 0.98$ [Proposed]	1.15 s	$\pm 0.82^\circ$	0.12°	Optimal balance; smooth yaw, zero drift
$\alpha = 0.99$	2.30 s	$\pm 0.45^\circ$	0.48°	Sluggish recovery during rapid switchbacks
$\alpha = 0.999$	5.80 s	$\pm 0.15^\circ$	2.85°	Excessive gyro drift; fails to track course

As synthesized in Table 5.4, an α value of 0.98 achieves the optimal trade-off, suppressing 91.3% of ambient compass noise while constraining gyroscopic bias drift to less than 0.12° per minute, ensuring ultra-stable heading tracking on complex forest trails.

5.2.7 Offline Vector Tiles vs. Raster Mapping Storage & Memory Benchmark

To quantify the storage and computational advantages of micro-sharded vector mapping, an empirical benchmark was conducted comparing TrailGuide's OpenStreetMap MVT/PBF protocol-buffer format against traditional pre-rendered PNG raster map tiles across zoom levels 13 through 18 over a 10 km x 10 km wilderness sector:

Table 5.5: Offline Vector Tiles vs. Raster Mapping Storage & Memory Benchmark

PERFORMANCE METRIC	CONVENTIONAL RASTER	TRAILGUIDE VECTOR	REDUCTION / GAIN	IMPACT ON FIELD OPERATIONS
Storage Footprint	485.6 MB	1.84 MB	99.62% Reduction	Enables offline storage of entire state parks
Offline Provision Time	420.0 seconds	1.6 seconds	99.61% Faster	Instant QR transfer at trailhead kiosk
Device Heap RAM	142.5 MB	18.5 MB	87.01% Reduction	Prevents OS low-memory app termination

Viewport Render Latency	38.5 ms / frame	8.2 ms / frame	78.70% Faster	Silky smooth 60 FPS vector zooming
Battery Drain / Hour	28.5% / hour	13.8% / hour	51.58% Conservation	Sustains full-day wilderness navigation

As detailed in Table 5.5, micro-sharded vector tiles achieve a 99.62% storage reduction over raster alternatives, collapsing a 485.6 MB raster footprint into a single 1.84 MB file that transfers over a QR scan code in 1.6 seconds. This drastically lowers mobile memory pressure and extends operational battery life by more than 50%.

5.2.8 Canopy Multipath Attenuation & BLE RSSI Log-Distance Modeling

Radio frequency propagation across forest vegetation was modeled using the empirical log-distance path loss formulation:

$$PL(d) = PL(d_0) + 10 \cdot n \cdot \log_{10}(d / d_0) + X_{\sigma} \quad (5.1)$$

Where $d_0 = 1.0$ m, $PL(d_0) = 42.5$ dBm at 2.4 GHz, and X_{σ} is zero-mean Gaussian shadow fading. Under dry foliage, the measured path loss exponent n was 2.85, whereas under heavy wet canopy conditions following rainfall, dielectric absorption increased n to 3.65 (compared to $n = 2.0$ in free space).

Despite this severe attenuation, TrailGuide's high-sensitivity BLE scanning (-95 dBm receiver sensitivity) sustains a single-hop communication radius of 75 to 90 meters beneath saturated multi-layered canopies. The Controlled Spray-and-Wait protocol successfully routes distress beacons through 5 successive hops over 450 meters within 4.2 seconds, confirming that opportunistic peer-to-peer forwarding provides a robust communication bridge during total cellular collapse.

5.3 COMPREHENSIVE DISCUSSION AND COMPARATIVE EVALUATION

To contextualize TrailGuide's contributions within the state-of-the-art of wilderness safety systems, Table 5.6 presents an architectural and operational comparison against commercial platforms and academic baselines:

Table 5.6: Comprehensive Wilderness Navigation & Emergency Architecture Comparison

FEATURE / METRIC	GARMIN INREACH	ALLTRAILS / GAIA	PURE SMARTPHONE INS	TRAILGUIDE [OURS]
Hardware Cost	\$400 - \$600 peripheral	\$0 (Smartphone app)	\$0 (Smartphone app)	\$0 (Commodity phone)

Monthly Subscription	\$15 - \$65 / month	\$35 / year (Pro plan)	\$0 (Free)	\$0 (Open source)
Canopy Positioning	Degraded (5-12m error)	Severe drift (>12.4m)	Unbounded drift (>15m)	Sub-meter (0.68m error)
Offline Map Package	Proprietary ROM	250 MB - 2 GB raster	None (No map engine)	< 2 MB vector tiles
Emergency SOS Radio	Iridium Satellite	Cellular Only (Fails)	None (No radio relay)	Peer BLE Mesh DTN
Gateway Cloud Dispatch	Garmin Response SAR	None	None	Next.js / Mongo SAR
Battery Endurance	Dedicated device	4 - 5 hours	6 - 7 hours	> 8 hours (13.8%/hr)

As synthesized in Table 5.6, TrailGuide achieves the high-reliability emergency beaconing of expensive satellite hardware and the sub-meter positioning of specialized inertial units while eliminating recurring subscription fees and hardware costs. By integrating low-overhead vector mapping, battery-conscious sensor fusion, and cloud-integrated rescue coordination, TrailGuide establishes an accessible, comprehensive off-grid wilderness safety net.

CHAPTER 6

CONCLUSION & FUTURE ENHANCEMENTS

6.1 CONCLUSION

This project presented TrailGuide, an autonomous multi-sensor fusion localization and emergency communication framework engineered for commodity smartphones operating in off-grid wilderness environments. By systematically addressing the catastrophic limitations of satellite canopy attenuation, unassisted inertial drift, raster map storage bloat, and communication blackouts, TrailGuide establishes an infrastructure-independent life-safety ecosystem.

The Sensor-Fusion Pedestrian Dead Reckoning (SF-PDR) engine successfully decouples positioning from degraded satellite signals under dense forest canopies. Fusing 100 Hz tri-axial accelerometer step kinematics, Weinberg dynamic stride length estimation, and a tilt-compensated complementary heading filter, the system achieves sub-meter horizontal accuracy (0.68 m mean error, 0.13 m drift per 100m) under 100% forest canopy occlusion, outperforming raw smartphone GNSS by 94.5% and 2019 GNSS+INS baselines by 86.0%.

The decentralized Bluetooth Low Energy (BLE) Delay-Tolerant Network (DTN) reliably propagates 24-byte emergency distress beacons across mobile peer nodes using a Controlled Spray-and-Wait protocol, delivering 100% packet success up to 2 hops and 97.5% reliability at 5 hops within 4.2 seconds. Relayed beacons are seamlessly ingested by the cloud-based Next.js and MongoDB Atlas Search and Rescue command center, enabling emergency responders to execute 2DSphere spatial queries and dispatch rescue teams via dynamic OSRM road routing.

Crucially, TrailGuide achieves these capabilities with zero external hardware costs, maintaining battery consumption at 13.8% per hour to ensure all-day off-grid operation. The framework delivers an accessible, robust technological safety net for outdoor adventurers, forestry researchers, and search-and-rescue teams.

6.2 FUTURE ENHANCEMENTS

Future extensions of the TrailGuide platform will expand capabilities in four strategic technical directions:

6.2.1 Long-Range Hybrid LoRa Mesh Bridging

Integrating low-power long-range (LoRa 868/915 MHz) transceivers into designated trail checkpoint relays and solar-powered wilderness beacons. By establishing a dual-radio BLE-LoRa hierarchy, localized mobile human clusters communicating over BLE can bridge distress packets across deep mountain valleys and inter-ridge distances exceeding 5 to 10 kilometers without requiring satellite

uplinks.

6.2.2 Barometric Pressure & Vertical Elevation Fusion

Incorporating smartphone MEMS barometric pressure sensor streams into the kinematic state estimator. Fusing atmospheric pressure differentials with terrain digital elevation models (DEM) will enable 3D pedestrian dead reckoning, accurately computing altitude gain, contour adherence, and multi-tier alpine climbing trajectories with sub-meter vertical precision.

6.2.3 On-Device Edge Neural Hazard Detection

Deploying highly quantized, lightweight convolutional neural network models (such as MobileNetV4-Small or YOLOv8-Nano) directly on mobile neural processing units. The model will analyze real-time camera frames during active trekking to automatically identify trail washed-out hazards, fallen timber blockages, landslide risks, and wildlife tracks, annotating hazards directly onto the shared offline vector map.

6.2.4 Multi-Agency Interoperability & Incident Reporting Standards

Extending the cloud SAR command portal to support international incident reporting standards, including Common Alerting Protocol (CAP v1.2) and Emergency Information Interoperability Framework (EIIF). This integration will enable direct telemetry synchronization with regional forest service dispatchers, national park rangers, and local disaster management agencies during large-scale search and rescue operations.

CHAPTER 7

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